

HQMoSS-Net: A Hybrid Quantum-Classical U-Net with Multi-Scale Quantum Mixture of Experts for Biomedical Image Segmentation

Anonymous

Anonymous Institution

Abstract. Accurate biomedical image segmentation remains difficult when annotated data are limited and target regions exhibit ambiguous boundaries, substantial scale variation, and spatially distant structures. We propose HQMoSS-Net, a parameter-efficient hybrid quantum-classical U-Net designed to address these challenges. Its key contribution is the integration of three complementary components: a Quantum Mixture of Experts that captures latent features through parameterized quantum experts operating at local, intermediate, and global scales; Residual Dilated Multi-Scale blocks that expand receptive fields and enrich contextual feature extraction in the encoder and decoder; and a Lightweight Selective Scan module for long-range spatial dependencies. Under a common five-fold cross-validation protocol, HQMoSS-Net achieves mean Soft IoU scores of 0.8497 on PH2 and 0.7038 on GlaS 2015, compared with 0.7787 and 0.5851 for U-Net, respectively. The model uses 153,660 trainable parameters and achieves higher overlap scores than U-Net and the evaluated classical U-Net variants under the same protocol. Component ablations indicate complementary improvements from the three components. These results demonstrate the effectiveness of HQMoSS-Net for biomedical image segmentation and motivate further research into improving the efficiency of deep neural networks through hybrid computational models.

Keywords: Biomedical image segmentation · hybrid quantum-classical learning · quantum machine learning · U-Net · mixture of experts · explainable AI

1 Introduction

Biomedical image segmentation identifies anatomical structures and pathological regions at the pixel level. U-Net [15] provides a foundational encoder-decoder architecture for this task, where skip connections combine semantic information with the spatial details needed for accurate segmentation. Building on this foundation, subsequent methods have incorporated residual learning [1], attention mechanisms [14], multi-scale feature processing [5], and long-range spatial dependency modeling [3]. Despite these advances, segmentation remains challenging because expert annotations are costly and scarce, while target regions

often have ambiguous boundaries, diverse sizes, or spatially disconnected components. Increasing model capacity can mitigate some of these limitations but also raises computational cost and the risk of overfitting on small datasets.

In recent years, quantum machine learning has attracted increasing attention for its ability to exploit quantum computational mechanisms for feature representation and processing. Quantum convolutional neural networks (QCNNs) [4] and quanvolutional neural networks (QNN) [7] are representative approaches. Due to current limitations of quantum computing, most studies focus on hybrid architectures in which quantum modules are integrated into classical deep networks. For image segmentation, QuNet uses a QuFeX block [10] at the U-Net bottleneck, whereas HQF-Net [8] employs a Quantum Mixture of Experts with adaptive routing for remote-sensing images. These studies establish a basis for combining direct quantum processing of feature patches with adaptive multi-expert routing in biomedical image segmentation.

Building on this basis, we propose HQMoSS-Net, a hybrid quantum–classical U-Net that combines direct quantum processing of feature patches with adaptive multi-expert routing at the bottleneck. The main contributions are:

- A QuantumMoE with three parameterized quantum experts that process latent feature patches at local, intermediate, and global extents. An adaptive pixel-wise router combines the expert outputs, while a residual connection preserves the input features.
- Integration of Residual Dilated Multi-Scale (RDMS) blocks into the encoder–decoder to expand receptive fields, together with a Lightweight Selective Scan (LSS) module at the bottleneck to model long-range spatial dependencies.
- A controlled evaluation using classical and quantum baselines, component and circuit-depth ablations, a matched Classical MoE control, and interpretability analysis with Seg-Grad-CAM [19].

HQMoSS-Net is evaluated through five-fold cross-validation on PH2 [11] and GlaS 2015 [17] under a common training protocol. The evaluation focuses on segmentation performance, parameter efficiency, the contribution of each component, and model interpretability.

The remainder of this paper is organized as follows. Section 2 reviews and analyzes related studies. Section 3 presents the HQMoSS-Net architecture and its constituent modules. Section 4 describes the experimental setup and analyzes the results. Section 5 summarizes the findings, discusses the study limitations, outlines future research directions, and concludes the paper.

2 Related Work

U-Net [15] forms the basis of many biomedical image segmentation methods through its encoder–decoder architecture and skip connections, which combine semantic information with spatial details. Subsequent variants introduced attention [14], residual learning [1], and dilated convolutions [5] to improve feature

representation and expand receptive fields. More recently, TransUNet [3] and VM-UNet [16] have exploited the Transformer [18] and Mamba [6], respectively, to model long-range spatial dependencies. However, the increasing complexity of these architectures can raise computational costs and the risk of overfitting on small biomedical datasets.

In quantum machine learning, QCNNs [4] and QNN [7] use quantum circuits to support feature extraction and representation. Given current limitations in qubit counts, circuit depth, and high-dimensional image encoding, hybrid quantum–classical architectures have received greater attention. QuFeX [10] directly processes latent feature-map patches and is integrated at the QuNet bottleneck. This approach preserves most of the classical U-Net structure, but relies on a single fixed quantum-processing path, which limits adaptation to different spatial extents.

Mixture of Experts (MoE) [9] combines multiple experts through an input-dependent routing mechanism. Following this principle, QMoE [13] uses parameterized quantum circuits as experts, but has been primarily evaluated on image-classification tasks. HQF-Net [8] extends Quantum MoE to remote-sensing segmentation with experts that process local, global, and directional information. However, it compresses features along both spatial and channel dimensions before quantum encoding, reducing the spatial information received by the quantum circuits. On this basis, HQMoSS-Net combines direct quantum processing of feature patches with adaptive multi-expert routing and uses RDMS and LSS to strengthen contextual representation. The model therefore addresses the identified gaps in combining quantum processing and multi-expert routing for biomedical image segmentation, while contributing to the exploration and application of MoE mechanisms in quantum machine learning, where such approaches remain relatively underexplored.

3 Proposed Method

3.1 Overall HQMoSS-Net Architecture

HQMoss-Net (Fig. 1) is a hybrid quantum–classical segmentation network based on a U-Net encoder–decoder architecture. The encoder uses the channel configuration $8 \rightarrow 16 \rightarrow 32 \rightarrow 16 \rightarrow 8$: the number of channels first increases to enrich feature representations and then decreases to produce a compact latent representation suitable for quantum processing while limiting the parameter count. The decoder restores the spatial resolution, while skip connections preserve localization information and boundary details.

RDMS is integrated into the encoder and decoder to extract features over different spatial extents. At the bottleneck, QuantumMoE processes patches with three quantum experts and combines their outputs through an adaptive router and a residual connection. LSS then models long-range spatial dependencies before the decoder produces the segmentation mask.

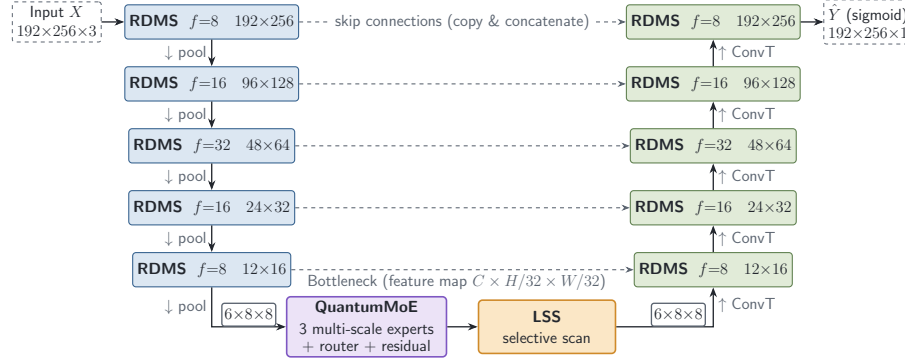


Fig. 1. Overall HQMoSS-Net architecture with RDMS encoder-decoder blocks and bottleneck QuantumMoE and LSS; dashed lines denote skip connections.

3.2 U-Net Backbone and RDMS Blocks

The classical part of HQMoSS-Net follows the U-Net architecture. The encoder progressively downsamples and transforms features under the predefined channel configuration, while the decoder restores resolution through transposed convolutions. Skip connections fuse features across the two paths to preserve localization information and boundary details. Building on this backbone, RDMS (shown in Fig. 2) replaces standard convolutional blocks to expand spatial context.

Given an input feature map x , a 3×3 convolution produces a shortcut feature s . Three depthwise dilated convolutions then process s in parallel with rates $r \in \{1, 2, 3\}$, providing effective receptive fields of 3×3 , 5×5 , and 7×7 :

$$b_r = \text{ReLU}\left(\text{DWConv}_{3 \times 3}^{(r)}(s)\right), \quad r \in \{1, 2, 3\}, \quad (1)$$

where $\text{DWConv}_{3 \times 3}^{(r)}$ denotes a depthwise convolution with dilation rate r , s is the projected shortcut feature, and b_r is the output of the corresponding branch. The three outputs are concatenated, projected by a 3×3 convolution, and added to s before the final ReLU activation:

$$\text{RDMS}(x) = \text{ReLU}(s + \text{Conv}_{3 \times 3}([b_1, b_2, b_3])). \quad (2)$$

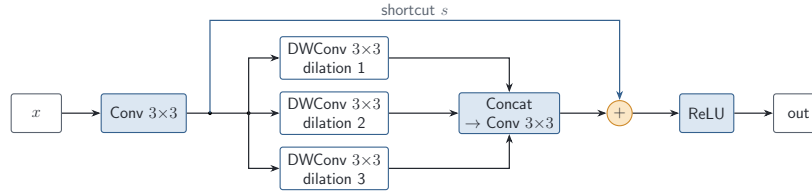


Fig. 2. RDMS block with three dilated depthwise branches and residual fusion.

3.3 Quantum Mixture-of-Experts at the Bottleneck

QuantumMoE is the main quantum module of HQMoSS-Net and is placed at the bottleneck to exploit compact latent features. Three quantum experts process patches at different spatial extents; their outputs are combined by an adaptive pixel-wise router and added to the input features through a residual connection. The module components are detailed in the following subsections.

Patch-based Quantum Experts Let $F \in \mathbb{R}^{B \times H \times W \times C}$ denote the bottleneck feature map, where B , H , W , and C are the batch size, height, width, and number of channels, respectively. Each quantum expert extracts overlapping patches of size $p_h \times p_w \times C$ with stride one and flattens them into vectors of length $N = p_h p_w C = 2^n$. Each vector is normalized using Layer Normalization (LayerNorm) [2], amplitude-encoded into n qubits, and transformed by a parameterized quantum circuit (PQC). Computational-basis measurement produces a probability vector of length 2^n , which is normalized, reshaped into a patch, and returned to its original location.

Because the patches overlap, the expert output at position p is obtained by averaging the corresponding processed values:

$$E(p) = \frac{1}{|\mathcal{N}(p)|} \sum_{(s,\delta): s+\delta=p} Q(s)_\delta, \quad (3)$$

where $E(p)$ is the expert output at position p , s is the starting position of a patch, δ is the relative offset of p within that patch, $Q(s)_\delta$ is the processed patch value at offset δ , and $\mathcal{N}(p)$ is the set of patches covering p . This averaging preserves the spatial dimensions of F ; the complete pipeline is illustrated in Fig. 3.

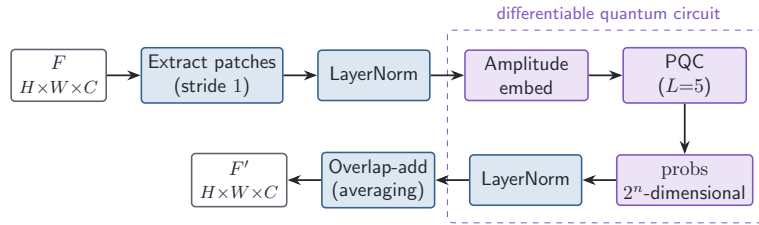


Fig. 3. Quantum expert pipeline with PQC processing and overlap-add reconstruction.

Quantum Circuit Design and Expert Configuration All three experts share the same PQC architecture but process patches with different sizes and numbers of qubits. After amplitude encoding, a Hadamard gate is applied to every qubit. The circuit consists of $L = 5$ trainable layers; within each layer,

each qubit is first applied a general rotation:

$$\text{Rot}(\phi, \theta, \omega) = \begin{bmatrix} e^{-i(\omega+\phi)/2} \cos \frac{\theta}{2} & -e^{-i(\omega-\phi)/2} \sin \frac{\theta}{2} \\ e^{i(\omega-\phi)/2} \sin \frac{\theta}{2} & e^{i(\omega+\phi)/2} \cos \frac{\theta}{2} \end{bmatrix}. \quad (4)$$

where ϕ , θ , and ω are trainable parameters; this rotation transforms a qubit about all three coordinate axes. After the rotation, the CNOT stage forms a ring by connecting adjacent qubits and linking the last qubit back to the first. The CNOT gate used for this connection is:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}. \quad (5)$$

At the final step of the circuit, R_X is applied to qubits satisfying $i \bmod 3 = 0$, while R_Y is applied to qubits satisfying $i \bmod 3 = 1$.

After processing, the circuit is measured in the computational basis. For the final state $|\psi\rangle$, the output probability vector is

$$\mathbf{p} = (|\langle 0|\psi\rangle|^2, \dots, |\langle 2^n - 1|\psi\rangle|^2) \in \mathbb{R}^{2^n}, \quad \sum_{j=0}^{2^n-1} p_j = 1. \quad (6)$$

where $|\psi\rangle$ is the final circuit state and p_j is the probability of measuring the j -th computational-basis state. Probability measurement retains the complete vector of length 2^n , rather than compressing the output into a few scalar values. The shared PQC and its measurement stage are illustrated in Fig. 4.

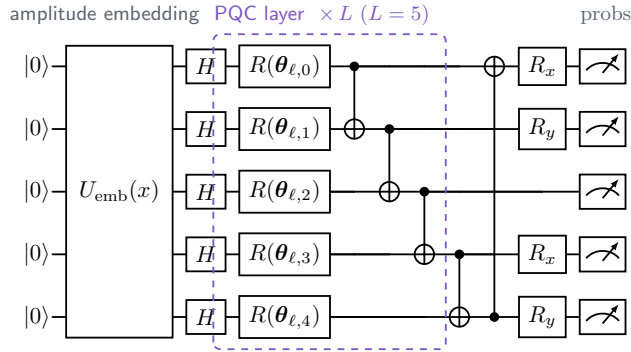


Fig. 4. Shared five-qubit PQC with ring entanglement and probability measurement.

The three experts process different spatial extents of the $6 \times 8 \times 8$ bottleneck feature map. For a patch of size $p_h \times p_w \times C$, the number of qubits is selected as

$n = \log_2(p_h p_w C)$, where p_h , p_w , and C are the patch height, width, and number of channels, respectively. The measured vector can therefore be reshaped directly to the original patch dimensions without an additional dimensional projection. The expert configurations are:

$$\begin{aligned} E_1 &: \text{patch } 2 \times 2, \quad 5 \text{ qubits}, \quad 5 \times 7 \text{ patch grid} \quad (\text{local}), \\ E_2 &: \text{patch } 2 \times 4, \quad 6 \text{ qubits}, \quad 5 \times 5 \text{ patch grid} \quad (\text{intermediate}), \\ E_3 &: \text{patch } 8 \times 8, \quad 9 \text{ qubits}, \quad 1 \times 1 \text{ patch grid} \quad (\text{global}). \end{aligned}$$

E_1 and E_2 process overlapping patches to extract local and intermediate features. For E_3 , the bottleneck feature map is padded from 6×8 to 8×8 and processed as a single patch to capture global context.

Router and Residual Fusion The spatial router concatenates the input feature F with the three expert outputs E_i , then uses a 1×1 convolution and softmax to predict position-wise weights:

$$(w_1, w_2, w_3) = \text{Softmax}(\text{Conv}_{1 \times 1}([F, E_1, E_2, E_3])), \quad \sum_{i=1}^3 w_i = 1. \quad (7)$$

where w_i is the weight assigned to expert i . The expert outputs are combined and added to the input through a residual connection:

$$\text{QuantumMoE}(F) = F + \sum_{i=1}^3 w_i \odot E_i. \quad (8)$$

Thus, the router adapts each expert’s contribution, while the residual path preserves the input representation.

3.4 Lightweight Selective Scan

The Lightweight Selective Scan (LSS) module complements QuantumMoE by modeling long-range spatial dependencies at the bottleneck. Inspired by selective state-space models [6], LSS applies LayerNorm and a 1×1 projection, then splits the features into a feature branch and a gate branch. The feature branch applies a 5×5 convolution and swish activation, flattens the $H \times W$ grid into a sequence, and scans it.

At each position in the sequence, the hidden state is updated as:

$$h_t = a \odot h_{t-1} + b_t \odot x_t, \quad (9)$$

where x_t is the input feature, h_t is the updated state, a is a learned transition parameter, and b_t is input-dependent. After scanning, the sequence is reshaped to the spatial grid, modulated by a sigmoid gate, projected to C channels, and added to the input through a residual connection. The output projection is

initialized with zero weights so that LSS starts close to an identity mapping and gradually learns long-range information. The module is illustrated in Fig. 5.

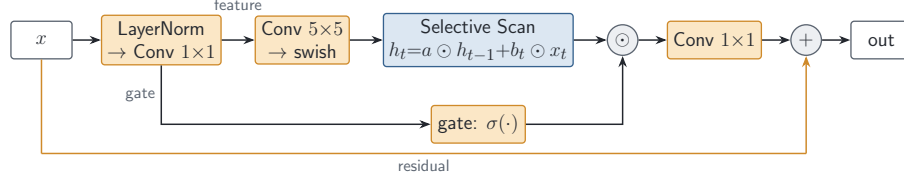


Fig. 5. LSS module with selective scanning, sigmoid gating, and residual connection.

4 Experiments and Results

4.1 Datasets and Experimental Setup

Experiments use two biomedical image segmentation datasets: PH2 contains 200 dermoscopic images with lesion segmentation masks [11], and GlaS 2015 contains 165 H&E-stained colorectal histology images with gland segmentation masks [17]. Images from both datasets are resized to 192×256 pixels.

All models are evaluated on each dataset using the same five-fold cross-validation splits (80/20 train/evaluation). The models are trained for 60 epochs with a batch size of 18, binary cross-entropy loss, and the Adam optimizer with a learning rate of 0.001. Experiments are performed on an NVIDIA Tesla T4 GPU, while the quantum circuits are simulated using PennyLane’s `default.qubit` backend. Results are reported as the mean and standard deviation over the five folds.

4.2 Comparison with Baselines

HQMoSS-Net is compared with five classical models: U-Net [15], UNet++ [20], Attention U-Net [14], V-Net [12], and R2U-Net [1], together with the quantum baseline QuNet using QuFeX [10]. Tables 1 and 2 report the mean $\pm$ standard deviation over five folds and the number of trainable parameters. Soft IoU is the primary metric because it evaluates predicted probabilities without binary thresholding.

HQMoSS-Net achieves the best result for all four metrics on both datasets. Its Soft IoU reaches 0.8497 on PH2, exceeding U-Net by 0.0710 and R2U-Net by 0.0439; on GlaS 2015, it reaches 0.7038, improving upon U-Net and Attention U-Net by 0.1187 and 0.0349, respectively. With 153,660 parameters, HQMoSS-Net is substantially smaller than most compared classical architectures while achieving higher scores. It uses more parameters than QuNet but consistently improves segmentation metrics on both datasets. These results show a favorable balance between segmentation quality and parameter efficiency.

Table 1. PH2 results over five folds (mean $\pm$ standard deviation; best in bold).

Model	Parameters	Soft IoU $\uparrow$	Hard IoU $\uparrow$	Soft Dice $\uparrow$	Hard Dice $\uparrow$
U-Net	88,209	0.7787 $\pm$ 0.0696	0.8422 $\pm$ 0.0191	0.8741 $\pm$ 0.0463	0.9142 $\pm$ 0.0112
UNet++	637,509	0.7928 $\pm$ 0.0284	0.8444 $\pm$ 0.0217	0.8842 $\pm$ 0.0178	0.9155 $\pm$ 0.0128
Attention U-Net	644,263	0.7923 $\pm$ 0.0358	0.8330 $\pm$ 0.0301	0.8837 $\pm$ 0.0227	0.9086 $\pm$ 0.0181
V-Net (2D)	1,110,933	0.7879 $\pm$ 0.0214	0.8259 $\pm$ 0.0306	0.8812 $\pm$ 0.0133	0.9044 $\pm$ 0.0182
R2U-Net	1,873,581	0.8058 $\pm$ 0.0277	0.8319 $\pm$ 0.0264	0.8923 $\pm$ 0.0171	0.9081 $\pm$ 0.0158
QuNet (QuFeX)	84,149	0.7942 $\pm$ 0.0177	0.8460 $\pm$ 0.0155	0.8852 $\pm$ 0.0109	0.9165 $\pm$ 0.0090
HQMoSS-Net	153,660	0.8497 $\pm$ 0.0196	0.8718 $\pm$ 0.0234	0.9187 $\pm$ 0.0116	0.9314 $\pm$ 0.0135

Table 2. GlaS 2015 results over five folds (mean $\pm$ standard deviation; best in bold).

Model	Parameters	Soft IoU $\uparrow$	Hard IoU $\uparrow$	Soft Dice $\uparrow$	Hard Dice $\uparrow$
U-Net	88,209	0.5851 $\pm$ 0.0620	0.6712 $\pm$ 0.0489	0.7367 $\pm$ 0.0496	0.8025 $\pm$ 0.0353
UNet++	637,509	0.6567 $\pm$ 0.1140	0.6789 $\pm$ 0.1070	0.7877 $\pm$ 0.0910	0.8046 $\pm$ 0.0820
Attention U-Net	644,263	0.6689 $\pm$ 0.0538	0.6971 $\pm$ 0.0600	0.8006 $\pm$ 0.0391	0.8203 $\pm$ 0.0425
V-Net (2D)	1,110,933	0.6187 $\pm$ 0.1475	0.6434 $\pm$ 0.1628	0.7559 $\pm$ 0.1175	0.7730 $\pm$ 0.1257
R2U-Net	1,873,581	0.6445 $\pm$ 0.0729	0.6712 $\pm$ 0.0813	0.7819 $\pm$ 0.0555	0.8010 $\pm$ 0.0593
QuNet (QuFeX)	84,149	0.6171 $\pm$ 0.0376	0.7033 $\pm$ 0.0388	0.7627 $\pm$ 0.0285	0.8254 $\pm$ 0.0262
HQMoSS-Net	153,660	0.7038 $\pm$ 0.0189	0.7498 $\pm$ 0.0191	0.8261 $\pm$ 0.0130	0.8569 $\pm$ 0.0125

4.3 Ablation Study

The ablation study is conducted on PH2 to assess the effect of circuit depth and the contribution of each component. Because many configurations are each evaluated using five-fold cross-validation, one representative dataset is used to control the cost of quantum simulation; the complete model remains evaluated on both PH2 and GlaS 2015.

As shown in Table 3, the (5, 5, 5) configuration achieves the highest mean across all four metrics and is therefore selected. Deeper circuits are not examined because simulation and gradient-computation costs increase with depth. In the five-layer control, Classical MoE has 1,693,908 parameters, 19.47 times more than the 86,996 parameter QuantumMoE, but yields a lower and less stable Soft IoU (0.7595 $\pm$ 0.1413 versus 0.8127 $\pm$ 0.0084).

Table 4 shows that each component improves Soft IoU over U-Net. RDMS gives the largest individual gain (6.18%), followed by QuantumMoE (4.37%) and LSS (1.90%). The complete model reaches 0.8497, corresponding to a 9.12% improvement over U-Net. QuNet improves by only 1.99%, indicating that the multi-expert mechanism and the integration of the components contribute substantially to HQMoSS-Net’s performance.

Table 3. Quantum-circuit depth ablation and a matched Classical MoE on PH2.

Layers per expert	Soft IoU $\uparrow$	Hard IoU $\uparrow$	Soft Dice $\uparrow$	Hard Dice $\uparrow$
Classical MoE (5 layers)	0.7595 $\pm$ 0.1413	0.8362 $\pm$ 0.0255	0.8567 $\pm$ 0.1022	0.9107 $\pm$ 0.0151
(1, 1, 1)	0.8103 $\pm$ 0.0271	0.8424 $\pm$ 0.0268	0.8950 $\pm$ 0.0166	0.9143 $\pm$ 0.0158
(2, 2, 2)	0.8105 $\pm$ 0.0174	0.8430 $\pm$ 0.0122	0.8953 $\pm$ 0.0107	0.9148 $\pm$ 0.0072
(3, 3, 3)	0.8077 $\pm$ 0.0391	0.8446 $\pm$ 0.0366	0.8932 $\pm$ 0.0243	0.9154 $\pm$ 0.0219
(4, 4, 4)	0.8089 $\pm$ 0.0210	0.8464 $\pm$ 0.0280	0.8943 $\pm$ 0.0128	0.9166 $\pm$ 0.0164
(5, 5, 5)	0.8127 $\pm$ 0.0084	0.8522 $\pm$ 0.0192	0.8967 $\pm$ 0.0051	0.9201 $\pm$ 0.0112

Table 4. Soft IoU improvements over the U-Net reference on PH2.

Configuration	Soft IoU	Δ absolute	Δ relative (%)
U-Net	0.7787	+0.0000	0.00
U-Net + LSS	0.7935	+0.0148	1.90
QuNet (QuFeX)	0.7942	+0.0155	1.99
QuantumMoE	0.8127	+0.0340	4.37
QuantumMoE + LSS	0.8258	+0.0471	6.05
U-Net + RDMS	0.8268	+0.0481	6.18
QuantumMoE + RDMS	0.8305	+0.0518	6.65
U-Net + LSS + RDMS	0.8340	+0.0553	7.10
HQMoSS-Net	0.8497	+0.0710	9.12

4.4 Qualitative and Interpretability Analysis

Beyond segmentation quality, we also assess the faithfulness and localization of the explanation maps using Deletion AUC, Insertion AUC, and the Energy Pointing Game; HD95 measures boundary accuracy. Lower Deletion AUC and HD95, and higher Insertion AUC and Energy Pointing Game, indicate better results. As interpretability analysis is complementary, the metrics in Table 5 are computed on the complete evaluation set of the first fold for both datasets.

Table 5. Interpretability and boundary quality of U-Net and HQMoSS-Net on the first-fold evaluation sets.

Metric	U-Net (PH2)	HQMoSS-Net (PH2)	U-Net (GlaS)	HQMoSS-Net (GlaS)
Deletion AUC ↓	0.2008 ± 0.2084	0.1756 ± 0.1982	0.5605 ± 0.1621	0.3688 ± 0.1346
Insertion AUC ↑	0.6379 ± 0.1020	0.7287 ± 0.1011	0.4878 ± 0.1538	0.4422 ± 0.0880
Energy Pointing Game ↑	0.9584 ± 0.0424	0.9864 ± 0.0175	0.7865 ± 0.1370	0.9393 ± 0.0589
HD95 ↓	9.3508 ± 7.7312	7.8690 ± 6.9155	49.5211 ± 38.2992	32.5693 ± 31.8151

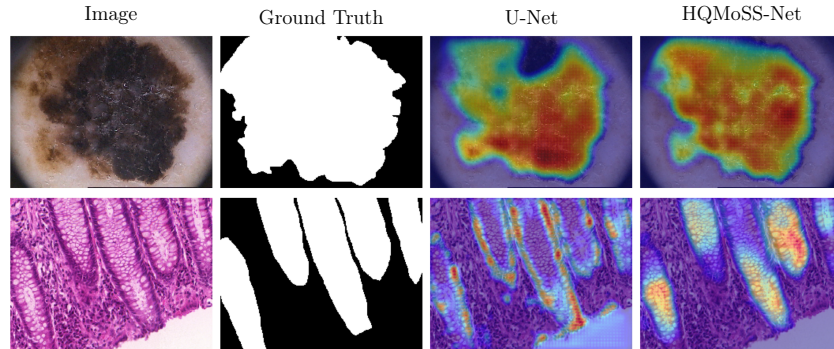
**Fig. 6.** Representative Seg-Grad-CAM maps on PH2 (top) and GlaS 2015 (bottom).

Table 5 shows that HQMoSS-Net improves Deletion AUC, Energy Pointing Game, and HD95 on both datasets. Notably, Deletion AUC on GlaS decreases

from 0.5605 to 0.3688, while HD95 on PH2 decreases from 9.3508 to 7.8690. Insertion AUC improves only on PH2, indicating that the results are not consistent across all metrics. In Fig. 6, the Seg-Grad-CAM maps [19] of HQMoSS-Net focus more clearly on the target regions and show less activation in the background than those of U-Net. These examples are illustrative and complement the quantitative analysis.

5 Conclusion

This paper proposed HQMoSS-Net, a hybrid quantum–classical U-Net integrating RDMS, QuantumMoE, and LSS for biomedical image segmentation. On PH2 and Glas 2015, the model achieves mean Soft IoU scores of 0.8497 and 0.7038, respectively, with 153,660 trainable parameters. Ablation analysis shows that the three components make complementary contributions: RDMS expands the receptive field, QuantumMoE enhances latent-feature processing, and LSS models long-range spatial dependencies.

In the matched control setting, Classical MoE uses 19.47 times more parameters but produces lower and less stable results than QuantumMoE. This finding suggests that the improvement is associated not only with the multi-expert organization but also with how QuantumMoE processes features.

Several limitations remain because the experiments use a noiseless simulator, evaluate only two relatively small datasets, and integrate quantum processing solely at the bottleneck. Moreover, a low parameter count does not imply low computational cost because quantum simulation and gradient computation remain expensive. Future work will consider more diverse datasets, noisy simulation, physical quantum hardware, and alternative locations for quantum integration. Overall, HQMoSS-Net demonstrates the potential of hybrid computational models to improve the effectiveness of biomedical image segmentation networks.

Competing Interests The authors declare no competing interests.

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